

2(a)	1 high / increased / better / not limiting, carbon dioxide / temperature / (named) mineral ions ; I nutrients 2 (from) coral / polyp, respiration / metabolism or for algae, Calvin cycle / light independent reactions ; A correct use of mineral ions	2
2(b)	<i>max 4 of:</i> 1 (paper / thin layer) chromatography / chromatogram ; 2 place spot of, extract / pigments, on pencil mark / at base of, paper / TLC plate ; 3 dry and repeat (to concentrate spot) ; 4 dip, paper / chromatogram, in solvent / so solvent travels up paper ; A named organic solvent (I water) R if spot submerged 5 measure distance travelled by solvent (front) and pigment (spot) ; 6 (calculate) R_f value = $\frac{\text{distance travelled by pigment}}{\text{distance travelled by solvent (front)}}$; 7 look up / compare results with, known R_f values (to identify pigments) ;	4
2(c)	<i>max 3 of:</i> 1 pigments absorb, violet-blue / 400–490 nm / lamp colours, well / best / most / at 8 out of 10 peaks ; 2 <u>rate of photosynthesis</u> of algae increases with more light absorbed ; 3 coral growth (increases) with more (algal) photosynthesis ; R products respond to give growth 4 chlorophyll a and peridinin are, most abundant pigments / most important ; 5 AVP ; e.g. violet-blue / 400–490 nm, predominate at the depths where corals live	3